

DATA SCIENTIST

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Summary:

3+ years of experience in Data Science, Machine learning, Python, and developing Chatbots, which provides solutions to business through research and in-depth analysis of data that helps in decision-making and different business domains.

Professional Skills:

- ❖ Worked on advanced analytical projects to reduce operation time, cost, and strategy teams within the different business domains.
- ❖ Leveraged **supervised and unsupervised machine learning algorithms** to build models and perform them, which gives automated solutions that reduce operational costs or improve operational efficiency in the industry.
- ❖ Applied analytical theory to real-world problems on dynamic datasets and performed hands-on to identify different data concerns. And getting data from different domains by using different web scraping techniques.
- ❖ To analyze the data, we used visualization techniques like univariate and **bivariate analysis through a statistical approach**.
- ❖ Improved the quality of data we have used **Preprocessing** steps like **Outlier treatment, Null value treatment, and Label Encoding techniques**.
- ❖ Competence in data visualization using **Matplotlib, Seaborn, and Plotly** and making it easy for stakeholders to understand in designing advanced dashboards to make decisions.
- ❖ Strong knowledge in creating analytical datasets using **Oracle SQL, queries, and joins**, and performed data operations. Had an experience in different deployment tools like AWS, Streamlit, and Flask.
- ❖ Leveraged data analytics and data science techniques to drive optimization, improvement of promotional strategies, and marketing techniques to assess the effectiveness
- ❖ Worked with stakeholders throughout the project in identifying the opportunities for leveraging company data to drive business solutions and coordinate with different functional teams to implement models and monitor outcomes.

Technical Skill Sets:

Algorithms:

- ❖ Linear Regression
- ❖ Logistic Regression
- ❖ K Means Clustering
- ❖ Agglomerative Clustering
- ❖ KNN Classification
- ❖ Decision Tree
- ❖ Random Forest
- ❖ SVM
- ❖ Naïve Bayes Classifier
- ❖ Boosting algorithms

Project Management:

- ❖ Jupyter Notebook
- ❖ Google Colab
- ❖ VS Code
- ❖ Git Hub

Python

Packages:

- ❖ SciKit Learn
- ❖ Numpy
- ❖ Scipy
- ❖ Pandas
- ❖ Stats model
- ❖ NLP
- ❖ RASA

Visualization Skills:

- ❖ Matplotlib
- ❖ Seaborn
- ❖ Plotly
- ❖ Microsoft office

Deployment Tools:

- ❖ Streamlit
- ❖ Flask
- ❖ AWS

Working Experience:

Company : Sunfra Software Services
Experience: 3.2 years
Role: Data Scientist

Career Experience:

Project Name **Reducing Customer Churn through predictive Analysis**

Tools **Python, SQL Server**

Techniques **Statistics, Logistic Regression**

Description:

- ❖ Client Company was one of the leading premium card subscription-based retailers. Every year customers will pay \$120 for a yearly subscription.
- ❖ As the new year approaches they would like to understand how many customers will not renew their cards for next month
- ❖ We have built a model that predicts whether the customer will churn.

Responsibilities:

- ❖ Responsible for collecting data from various sources such as customer demographics, transactional data, and credit history. The data will be preprocessed to remove any inconsistencies and missing values.
- ❖ Performed exploratory data analysis (EDA) to identify patterns and trends in the data. Visualizations such as histograms, scatterplots, and heat maps will be used to understand the relationships between variables.
- ❖ Handle imbalanced datasets with the help of Smote technique. Responsible for designing test and train data sets and executing supervised machine learning algorithms such as logistic regression, decision trees, and random forests will be used to develop the classification model. The model will be trained using historical data and validated using a holdout set of data to ensure its accuracy
- ❖ Analyzed the classification report, accuracy score, and ROC curve to gauge the performance of the models.
- ❖ Identified the TPR and FPR for the models and communicated the KPIs to the stakeholders.
- ❖ Conducted feature selection Feature selection techniques such as correlation analysis, feature importance, and stepwise regression will be used to identify the most important features that impact customer churn.
- ❖ Increase the model accuracy by performing multicollinearity, and variance inflation factor analysis.
- ❖ Used K folded technique to compare the accuracies across the models and pushed the results of the best models.

Project Name **Sentiment Analysis of Customer Reviews.**

Environment/Technology Python, SQL, scikit-learn, NLTK Algorithms: Naïve Bayes, Random Forest, TF-IDF.

Description:

- ❖ This project focuses on analyzing customer reviews in the retail industry to extract sentiment and gain insights into customer satisfaction. By applying natural language processing (NLP) techniques, the project aims to automatically classify customer reviews as positive, negative, or neutral. The sentiment analysis results will help retailers identify areas for improvement and enhance the overall customer experience.

Roles and Responsibilities:

- ❖ Collect and preprocess customer review data, removing noise and irrelevant information.
- ❖ Apply NLP techniques to tokenize, clean, and represent text data.
- ❖ Train and fine-tune text classification models for sentiment analysis.
- ❖ Evaluate model performance using appropriate metrics and iterate on the models to improve accuracy.
- ❖ Generate insights and visualizations from the sentiment analysis results to guide business decisions and improve customer satisfaction

Project Name **Specifying the top-performing Warehouses across specific Geographies.**

Environment/Technology **Oracle SQL, Python**

Methodology **K Means Clustering, Agglomerative Clustering**

Responsibilities:

- ❖ Raw data Extraction, data cleansing, validation of data and Data reformat using Excel as per the client's specification.
- ❖ Created SQL datasets and Descriptive Statistics analysis by using Python.
- ❖ Cluster Analysis for grouping the shop type (Market Segmentation) which gives similar characteristics.
- ❖ Expertise in Analyzing and giving insights into the client's queries by using statistical procedures like Means, Univariate, etc.
- ❖ Providing insights using retail sales data from the previous and current periods to determine the patterns and trends.
- ❖ Data management activities like merging, appending datasets, handling missing values reading raw data files, handling programming errors, and accessing and managing data.
- ❖ Providing weekly and monthly reports as per the client's requirement.
- ❖ Classifying variables into different segments basis various customer variables like: - Wallet size, website visit frequency, time spent on the website, Product views, etc.
- ❖ Coming up with meaningful customer segments that were given to CRM and social channels like Facebook and LinkedIn for targeting them with relevant offers and increase the revenue per customer

- ❖ Identifying customer basis parameters like: - last transaction date, number of returned orders, number of calls, etc., that have the highest probability of not returning to the website in the next 90 days.

Project Name **Face emotion detection**

Environment/Technology **Python, OpenCV ,YOLOv8, YOLOs ,Deep learning.**

Description:

Spearheaded a project focused on real-time facial emotion detection using state of the art computer vision techniques, specifically YOLOv8 and YOLOs (You Only Look Once) models. Successfully developed a system that accurately detects and classifies facial emotions in images and video streams, contributing to applications in customer sentiment analysis and market research.

Responsibilities:

- ❖ Researched and selected appropriate pre-trained YOLOv8 and YOLOs models for object detection.
- ❖ Designed and implemented a custom dataset by collecting and annotating facial images depicting various emotions.
- ❖ Pre processed the dataset, including image resizing, augmentation, and normalization, to improve model performance.
- ❖ Fine-tuned the selected YOLO models using transfer learning on the emotion dataset.
- ❖ Integrated OpenCV to capture real-time video streams from webcams and process frames for emotion detection.
- ❖ Developed a user-friendly interface to display real-time video with emotion labels overlay.
- ❖ Conducted extensive testing to evaluate the accuracy and robustness of the models across diverse scenarios.
- ❖ Optimized model parameters and hyperparameters to achieve higher accuracy and faster processing speed.
- ❖ Collaborated with UX/UI team to design intuitive visuals for the interface, enhancing user experience.
- ❖ Documented the entire process, including data collection, model selection, training strategies, and technical details.

Educational Qualification:

Universities/Institutes:

NBKR Institute of Science and Technology, Andhra Pradesh.

Qualification: Bachelor of Technology (2019).

Declaration:

I hereby declare that the information furnished above is true and correct to the best of my knowledge.

Place: - Bangalore

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